Shinhan Soft POS

The Merchant Operating System with Embedded Credit

Shinhan Future's Lab Hackathon

The Problem

Vietnamese SME Merchants Struggle Every Day

Daily Pain Points:

- Managing POS, Shopee, TikTok, MoMo separately
- Counting inventory manually every evening
- Tracking staff hours and shift schedules
- Reconciling payments from 5+ platforms
- Cash flow gaps between sales and settlements

Current Reality:

- 3 different apps for sales channels
- Excel spreadsheets for inventory
- Paper timesheets for staff
- 2+ hours nightly reconciliation
- No visibility when cash will run out

The Credit Gap

What Banks Want	What SMEs Have
6 months statements	Cash registers (POS)
Tax returns	Shopee/Lazada/TikTok sales
Collateral	MoMo wallet transactions
Formal business registration	Daily sales receipts

40.5% of Vietnam's firms have bank loans — World Bank Enterprise Survey

The Solution

Shinhan Soft POS

The Alternative Credit Layer for Vietnam's SMEs

✅ POS + marketplace + e-wallet data fusion
✅ ML-based normalization turns fragmented activity into one merchant timeline
✅ Alternative credit profile built from explainable transaction patterns
✅ Signals banks can actually underwrite against

✅ Embedded working capital at the point of need
✅ 72-hour approval target with policy-based pre-screening
✅ No traditional bureau file required
✅ Human-reviewed underwriting inside daily utility

Daily Utility

This is what merchants do EVERY DAY

🌅 **Morning**
Check overnight orders
Review inventory levels
Assign staff shifts

☀️ **Midday**
Process in-store sales
Fulfill marketplace orders
Print customer receipts

🌇 **Evening**
Auto-reconcile all payments
See low-stock alerts
Close shift reports

📈 **Any Time**
Check sales performance
Manage staff
Restock supplies

Engagement Thesis

Why merchants come back every single day

❌ This is NOT a loan app

Merchants don't install loan apps. They install tools that run their business.

✅ This is their operating system

- Their business lives inside this app
- Trust is built through daily usage
- Transaction data accumulates naturally
- Financing appears only when they need it

Embedded Capital

Financing when it matters, not as an ad

Contextual Triggers:

- 📦 Low stock alert → **Restock financing**
- ⏳ Settlement delayed → **Cash flow advance**
- 📈 Sales growing fast → **Growth capital**

How it works:

- Based on real business performance
- No separate application
- No credit check required
- Fast, transparent, fair terms

The Loan Product

Clean loan terms for eligible merchants

Loan Range	**$200 - $5,000**
Interest Rate	15-18% APR
Term Length	3, 6, or 12 months
Approval Time	72 hours
Min. History	90 days

Demo Flow

How merchant activity turns into financing

Step 1

Install SOL POS

Merchant starts using the operating tool every day.

→

Step 2

Data Normalization

Transactions and consented mobile or telco signals are cleaned and standardized.

→

Step 3

Feature Profile Forms

Audit-ready transaction features populate the scorecard profile.

→

Step 4

90 Days of History

Seasonality, stability, and risk features become reliable.

→

Step 5

Policy Eligibility

Policy rules + score threshold determine eligibility for underwriter review.

→

Step 6

Loan Offer Shown

Illustrative amount and terms appear inside the workflow.

→

Step 7

Funds in 72 Hours

Approved merchants receive funds after underwriting and policy checks.

AI Governance Framework

Explainable scoring with human review by default

Human-in-the-Loop as Default

- Every score/recommendation → human review
- Fixed reason codes accompany each output
- LLM translates only approved reason codes
- Inputs, weights, bins logged for audit

What AI Actually Does (scoring path)

- Transaction cleaning & categorization
- Merchant behavior feature engineering
- WoE logistic scorecard → primary model
- EBM challenger → validation only

What AI Does Not Do

- Final credit decisions (bankers decide)
- Custom explanations (approved codes only)
- Black-box inference in scoring path

Explainable Credit Intelligence Layer

Where Shinhan's underwriting logic stays transparent

ML-Based Data Normalization

POS, marketplace, e-wallet & telco metadata fused into one audit-ready merchant timeline.

WoE Scorecard Core

Fixed bins, weights & reason codes. EBM challenger runs in parallel for accuracy check.

Statistical Pattern Features

Tết, monsoon & household-business effects → peer-normalized features inside scorecard.

AI-Assisted Underwriting

Policy rules + score thresholds → recommendation pack. Human review standard. LLM translates approved codes only.

Every score is reproducible from logged inputs, weights, bin boundaries, and reason codes.

The Data Flywheel

01
More merchants
start using the system

02
More data fused
through consented data normalization

03
Scorecards improve
through calibration and challenger validation

04
Faster underwriting
for strong merchants

05
Better risk features
add seasonality, stability, and anomaly context

06
Lower defaults
through stronger risk selection

07
Better portfolio
performance for lenders

08
Lower cost of capital
from stronger confidence

09
Lower rates
create more demand

Performance data improves the next validation cycle

Competitive Advantage

	Traditional POS	Marketplace Apps	Bank Loans	Shinhan Soft POS
Multi-channel orders	❌	✅ Only own channel	❌	✅ **All channels unified**
Inventory management	⚠️ Basic	❌	❌	✅ **Smart alerts**
Staff management	❌	❌	❌	✅
Automatic reconciliation	❌	❌	❌	✅
Contextual financing	❌	❌	❌	✅ **Embedded**
Daily usage frequency	Daily	Weekly	Monthly	**Multiple times per day**

Sources: MISA Lending (misa.vn), Shopee SEasy (TPBank partnership), Validus Vietnam, Techcombank ShopCash

Why This Matters:

- Existing solutions use ONE data channel only
- Our approach combines POS + Marketplace + E-wallet data — not yet disclosed as a combined approach in Vietnam

Payment Safety

How we protect merchants and Shinhan

3 statistical safety patterns we monitor:

🔄

Circular Wallet Flows

Rule-based flow checks flag repeated wallet loops that appear to inflate transaction volume.

🛒

Synthetic Orders

Cross-checks compare payment, address, and fulfillment trails to surface synthetic orders.

⚡

Velocity Anomalies

Threshold and peer-baseline checks surface unusual spikes at odd hours or from new devices.

Vietnam-Specific Risk Signals

Peer-normalized statistical signals tuned to how local merchants actually operate:

🎊

Tết Season Patterns

Lunar-calendar seasonality factors help prevent holiday spikes from being mistaken for permanent growth.

🌧️

Monsoon Disruption

Regional baseline adjustments separate weather-driven demand drops from true merchant deterioration.

👪

Family Business Structures

Linked-counterparty features capture shared household cashflow patterns while keeping the score fully auditable.

Compliance Roadmap

Path to Production

Phase 1

Sandbox (Month 1-2)

- Anonymized data only
- Internal testing
- Score + reason codes prepared for analyst review
- SBV regulatory consultation

→

Phase 2

Pilot (Month 3-4)

- Live data with full consent
- POS operations first
- Automated score calculation and policy checks
- Human review remains standard
- Compliance audit trail

→

Phase 3

Production (Month 5-6)

- Full POS rollout
- Contextual capital offers
- Explainable scorecard + audit trail
- Human-reviewed origination at 100+ merchants

The Ask

What we need

- Access to merchant/payment infrastructure
- Partnership with existing POS providers
- Pilot merchant cohort for validation
- **6 months** to prove daily engagement

What you get

- The daily merchant operating system for Vietnam
- High frequency merchant relationship
- Embedded financing monetization layer
- Data asset that compounds with every transaction

What This Is

We haven't launched a production credit product yet.

We don't have live API integrations yet.

We don't have a backtested model yet.

What we HAVE built:

- Working early system showing normalized data → explainable score concept
- ML-based data fusion and scorecard pipeline architecture
- Clear path to 6-month early product launch

This is an early product proposal with a defined roadmap.

The Close

Every sale is captured.

Every inventory alert is acted on.

Every merchant grows with Shinhan.

Position Shinhan as the bank merchants use every day.

Appendix: Research Sources

Market Context

Source	What It Verifies
Vietnam News - Digital Loans Open New Doors for SMEs	SMEs struggle to access bank credit; digital lending growing
Techcombank - MISA Lending	Existing bank alternative-data lending product
MISA Lending - MB Partnership	Digital SME lending via accounting software

Existing Competitor Products

Source	What It Verifies
Shopee Seller Education - SEasy	Shopee seller financing product exists
Validus Vietnam	Active SME lender in Vietnam
Techcombank MerchantOne	Bank merchant acceptance product
Techcombank SoftPOS	Bank SoftPOS product
Techcombank ShopCash	Bank merchant lending product

Appendix: Model Explainability

WoE Scorecard & EBM Challenger

WoE Logistic Scorecard

Converts transaction history into a single score (300–850 scale) with fixed reason codes. Every input, weight, and bin boundary is logged and reproducible. Regulators prefer it because every decision is fully auditable.

EBM Challenger Model

A glass-box ML model that runs alongside the scorecard for accuracy validation. Each feature's contribution to the final score is visible. Used for validation only — not final decisions.

Why Not Deep Learning or LLM for Scoring?

> Black-box models are prohibited by most banking regulators — no one can explain why a merchant was declined. Our LLM is limited to translating pre-approved reason codes into natural language. It never makes or modifies credit decisions.

Every score is reproducible from logged inputs, weights, bin boundaries, and reason codes.